

355 Python Programming

Description & Eligibility

Contestants will evaluate knowledge of working with structured designs, algorithms, and OOP methodology using standard libraries in the Python programming language. This event will be graded on the Windows operating system; therefore, Unix/Linux should not be used. Any secondary or postsecondary division contestant may enter this application event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Computer or laptop/notebook (USB-Type A or adapter required); a full-size keyboard may be used (no printer is needed) and supporting equipment (extension cords, etc.)
- Contestant must use one of these IDEs: IDLE, Visual Studio, Visual Studio Code, PyCharm, VSCode
- Only pre-written code that is hard copied is allowed

Contest Notes

- Method of evaluation: application
- AI Policy Level 1 *No AI Allowed*
- Carry-in and setup of equipment must be done solely by the contestant within the time allotted
- All student-produced jobs completed on-site are uploaded electronically and are not printed at the national level
- Business Professionals of America assumes no responsibility for hardware/software provided by the contestant
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. Violation of this rule will result in disqualification
- At NLC, contestants will take IT Specialist - Python certification as part of the event. Contestants passing this certification will have 50 points added to their event score.
 - A Pearson profile must be created at <https://certiport.pearsonvue.com> before attending NLC
 - Contestants must include their BPA Contestant ID in their Pearson profile when they register
 - Certiport Parent/Legal Guardian Consent Form must be uploaded to: education@bpa.org
 - For more information on the exam, visit <https://certiport.pearsonvue.com>
 - Reference materials are not allowed for certification testing
- Length of event: no more than ten (10) minutes orientation, no more than ninety (90) minutes testing time, and no more than ten (10) minutes wrap-up
 - No more than fifty (50) minutes for certification test
- Entries vary by state

Contest Competencies

1. Demonstrate proficiency in Python programming and apply algorithmic thinking, language syntax, core concepts, and modern practices to implement effective Python solutions
2. Develop modular, maintainable programs using foundational object-oriented principles in Python
3. Use Python file I/O to read, write, and manage data in applications
4. Apply structured and object-oriented design techniques and implement effective error-handling strategies to create scalable, maintainable, and resilient Python applications
5. Create and interpret UML diagrams to model software architecture and communicate design decisions in Python projects
6. Utilize Python's built-in collections and data structures—including lists, sets, dictionaries, tuples, iterators, or related utilities—to efficiently store, manage, and manipulate data
7. Apply standard Python libraries and modern language features to write clean, readable, and maintainable code